Assistive Technology for Hearing-Impaired and Deaf Students Utilizing Augmented Reality

Ali Mohammed Ridha[1,2], Wessam Shehieb[1,2]

[1] School of Electrical & Electronic Engineering, Universiti Sains Malaysia, Penang, Malaysia
[2] Department of Electrical and Computer Engineering, Ajman University, Ajman, U.A.E
Email: ali.noman.moh@gmail.com, w.shehieb@ajman.ac.ae

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***Abstract*— Common universities and colleges worldwide are not equipped with technological techniques that can assist deaf and hearing-impaired students in their education, while few specialized universities are built for deaf students, sign language tutors, or in-class television captioning are included for few majors only. A novel system is proposed and developed to help deaf and hearing-impaired students who are aiming to continue their education outside special centers and communicate normally with the societies that are not sign language educated. By introducing an intelligent software solution developed for affordable augmented reality glasses that will assist students in their educational journey with real-time transcribing, speech emotion recognition, sound indications features, as well as classroom assistive tools.**

Keywords*— *Assistive technology; Augmented Reality; Deaf; Education; Hearing-Impairment; Machine Learning.

I. Introduction

One of the main impacts of Hearing impairment and deafness is the ability to communicate effectively with others, which can have significant adverse effects on children's development and performance [1]. It is estimated that more than 430 million people have disabling hearing loss including 34 million children worldwide, and it's projected that 2.5 billion people are at risk of hearing loss by 2050 [2]. While education is an essential aspect of a child's development, Hearing-impaired and deaf (HID) students often have increased rates of grade failures and greater needs for education assistance [3], but the lack of adequate educational institutes that fully supports HID students' education can have a negative impact on their development. According to Article 26 of the United Nations Universal Declaration of Human Rights [4], it is necessary to allow HID students to integrate into any educational institute with education being a human right.

Current methods to teach HID students are to provide sign language instructors or in-class captioning systems [5], but these solutions can be expensive and difficult to manage, as well as having the students losing important visual gestures and motions of the instructor by limiting their focus on the interpreter.

In a short survey made in the United Arab Emirates targeting school and university students as shown in Figure 1, the majority of the students have never encountered any HID students in their educational journey. Additionally, they agree with the need for new technology to integrate them into mainstream education. It is interpreted that the reason most educational systems are not equipped with the technological techniques needed for HID students due to the complexity of the assessment procedure and staff expenses, instead HID students attend schools designed especially for them, and in many scenarios, they never attend universities, causing them to be isolated from the society we live in.

Modern Assistive technologies that are being researched and employed, make use of virtual reality, Augmented reality, and speech captioning as a means to assist HID students in learning [6]. While most of the current research focuses on in-class assessment techniques and evaluation [7,8], the integration of the mixed reality opened the door to enhance their experience a bit further than the typical methods. Other researchers [9] proposed a more effective student-to- teacher experience with an after-class learning care platform using smartphones. The previously developed research [10] is a system that can diagnose hearing impairment and earwax blockage which is capable of indicating the severity of the condition using a percentile scale via a smartphone application, to be incorporated in schools and other health awareness campaigns.

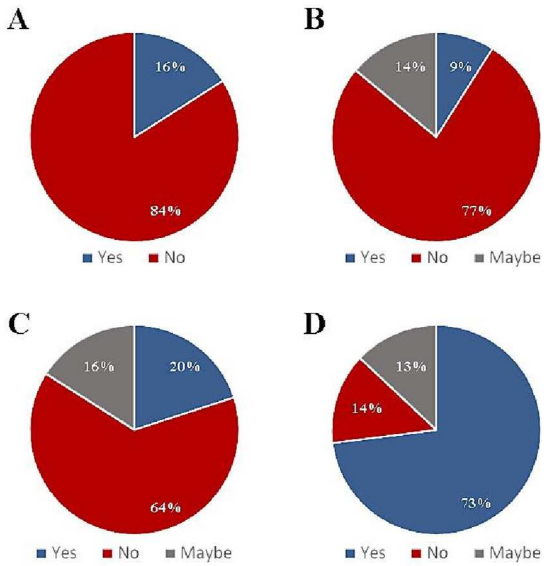


Figure 1: A survey about students' realization of HID students' status in the educational sector with a total of 109 participants.; (A) Have you met someone with a hearing disability studying in a school or university? (B) Is your university equipped with the technology to support? (C) Do you think people with hearing disabilities have good higher education? (D) Do you think there is a need for a device that can allow students with hearing disabilities to join mainstream education?

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However, all the current research is lacking the important aspect of integrating the students into society, as these solutions are employed in educational institutes only and focusing mainly on assessment, not inclusiveness.

In this work, a novel intelligent approach is proposed, that will allow HID students to join any desirable educational institution to pursue their dreams without any limitations. In addition to improving their communication skills with their colleagues. The proposed project will introduce a compact solution that will empower HID students in the community and allow easier access to education on any level.

II. Design and Implementation

Using Augmented reality with smart glasses was the most convenient approach to assist the patient with hearing disabilities without standing out and to provide them with direct access (live) to the features. The system presented in this paper will have multiple main features that rely on machine learning algorithms working in conjunction with each other continuously. The system was initially developed on MadGaze X5 smart glasses [11] as shown in Figure 2 with an attached microphone array system connected to the glasses.

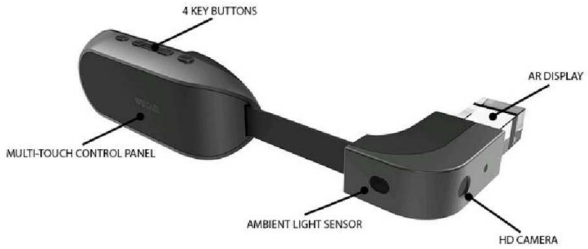


Figure 2: MadGaze X5 glasses [11].

The proposed final prototype design which will be used in future work is shown in Figure 3, A novel augmented reality glasses with Android operating system that has the appropriate processing power to have the developed software run flawlessly, and it will have a built-in microphone array covering all sides, the augmented reality will only be in one lens, and the other side can be a normal lens or even a prescription lens.

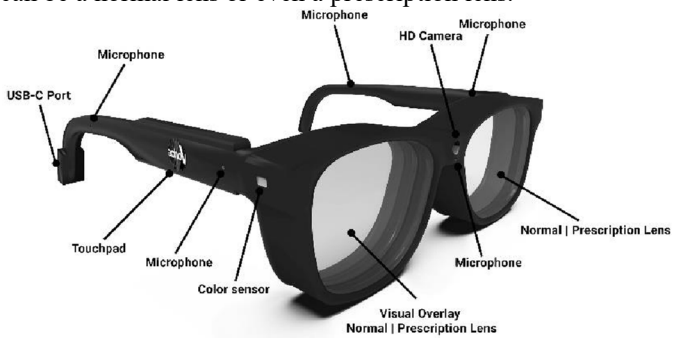


Figure 3: Proposed prototype design.

The complete glasses prototype shown in Figure 4 will offer four features: Real-time speech to text subtitles, Speech emotion recognition, environmental sounds indications, and classroom assistive tools. Each of these features is explained in more detail with their obtained results in the following parts of the paper.

The prototype provides a variety of features by utilizing a selection of hardware and software.



Figure 4: AR glasses prototype.

A. Hardware Design and Implementation

The hardware segment of the system consists of the Smart glasses and the Microphone array connected to the glasses through USB. The microphone array consists of a teensy 3.1 microcontroller chosen for its speed and small size, and 4 MEMS microphones [12] as shown in Figure 5, the PCB schematic, the teensy is used to process the microphone array data and send the location of surrounding environmental sounds.

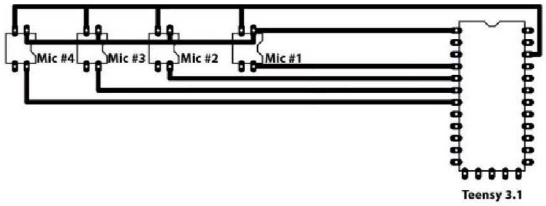


Figure 5: PCB Schematic.

B. Software Design and Implementation

The application was developed using Android studio [13], and has employed several libraries such as Firebase, and Tensorflow [14]. To provide the patients with one convenient application that will offer all four features, as seen in Figure. 7, the application layout tree.

III. Test Results

The four main features are primarily designed to tackle the problems deaf and hearing-impaired students face, the features were first developed on an Android smartphone, then later to the Android AR glasses.

• *Real-time speech to text subtitles (RSPT)* The RSPT feature will convert the incoming speech to text and include multiple colors for identifying different speakers (Speaker diarisation), the feature uses Google Cloud Speechto-text API [15] and will function as shown in Figure 6.

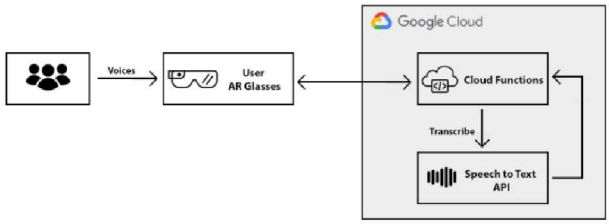


Figure 6: Real-time speech to text subtitles (RSPT) feature Block Diagram.

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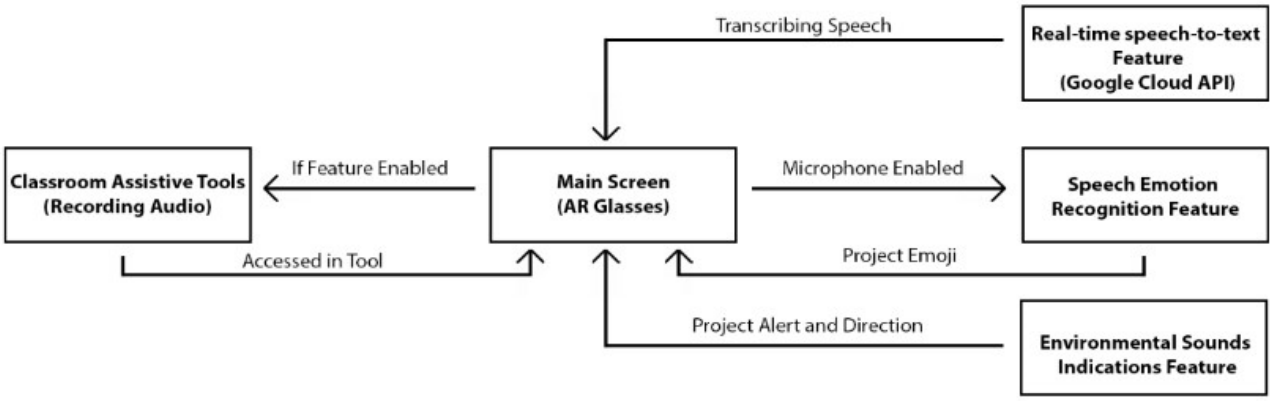


Figure 7: Application layout tree.

• *Speech Emotion Recognition (SER)*

The Emotion Recognition feature will use speech to predict the emotional state of the speaker, it utilizes a Convolutional neural network Machine learning model that is trained on two opensource datasets, The Ryerson Audio-Visual Database of Emotional Speech and Song [16] and Berlin Database of Emotional Speech [17], The predicted emotion then will be projected on the screen using an emoji as shown in Figure 8. The model achieved a validation accuracy of about 71.3%, as seen in Table I, some sample predictions. and the model should be able to perform better, with a bigger dataset.

Table I: Sample Predictions for SER

Input Audio	Predicted	Predicted Value
Fearful	Happy	0.62
Fearful	Fearful	0.87
Sad	Sad	0.73
Angry	Fearful	0.66



Figure 8: Speech Emotion Recognition (SER) as projected on the AR Glasses.

• *Environmental Sounds Indication (ESI)*

This feature will be utilizing the mems microphone array, and it will identify the surrounding sounds patterns such as broken glass, car horn or even gunshots, it will be a Convolutional neural network trained on the UrbanSound dataset [18], plus a dataset of collected sounds, it will predict the sound and position then project it on the AR screen as shown in Figure 9.



Figure 9: Environmental Sounds Indication (ESI) as projected on the AR Glasses.

The model achieved a validation accuracy of approximately 52.5%, as seen in Table II, some sample predictions.

Table II: Sample Predictions for ESI

Actual	Predicted	Predicted Value
Car Horn	Car Horn	0.64
Siren	Car Horn	0.56
Gun Shot	Gun Shot	0.77
Broken Glass	Broken Glass	0.57

• *Classroom Assistive Tools*

A collection of features that enables the student to have the best class experience and assist them in their daily tasks, it includes a notes recorder for lectures, which will record and transcribe the lecture for later use (with the consent of the educational institute) as shown in Figure 10, it will also include a collaboration platform to share notes as well as an exam mode, that disables the use of such features for secure examination.

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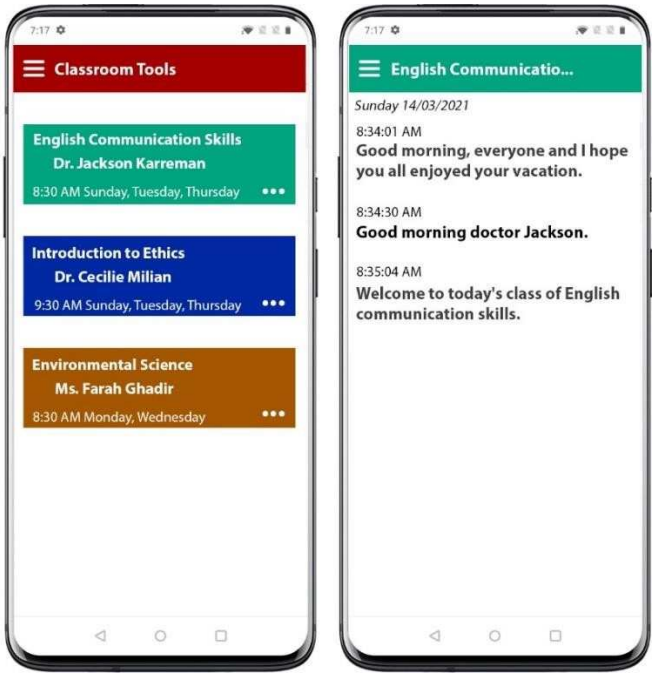


Figure 10: Classroom Assistive Tools smartphone application linked with the system.

IV. Discussion

The current developed system is fully functional but has yet to be used by an actual HID student, therefore; there could be still some room for improvements, in its comfortability by integrating the microphone array inside the AR glasses without it being an external bulky component. Another aspect that can be improved is the accuracy of the features by collecting new data and re-training the neural network models to increase the system's overall performance.

V. Conclusion

This paper proposed and developed a novel solution for students suffering from hearing-impairment or deafness to give them the freedom to choose any educational institute to pursue their dreams without any additional facilities needed. The student will be able to understand the lectures with the live transcription feature that uses Google Cloud services, additionally storing the transcribed lectures for future reference in the classroom tools feature, that can also be shared among other students, making it a platform that can be used for communication between students. The system will also allow the user to read the emotional state of the person talking by the tone of their voice, which will help the user to have the real experience of conversing with people. Furthermore, the environmental sounds indication feature would assist the user in their everyday life to have an understanding of their surroundings. Therefore, the developed system can be an extremely practical and useful tool not only to students but also to any individual suffering from hearing impairment or deafness to assist them in their everyday lives.

VI. Future Work

Since the system was built on ready-made AR glasses, with the addition of a developed microphone array, the development of an AR glasses that includes everything that is needed to give the optimal performance is still required, to balance between many factors including users' comfort, glasses size, and processing power, to create the best possible user experience.

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